

Secure Shell (SSH) Theory (1 hour):

Importance of SSH for secure communication. Configuring and securing SSH.

Practical (1 hour): Configure SSH with key-based authentication. Task: Disable password login and document the configuration.

TASKS

1. Install the OpenSSH server package on your Linux system and verify that the sshd service is running by connecting to localhost using the ssh command.

Install OpenSSH and Verify

Step 1: Install OpenSSH Server

```
dnf install openssh-server -y
```

Step 2: Start SSH Service

```
[root@localhost ~]# dnf install openssh-server -y
CentOS Stream 9 - BaseOS                3.9 kB/s | 4.8 kB      00:01
CentOS Stream 9 - AppStream              4.0 kB/s | 5.5 kB      00:01
CentOS Stream 9 - Extras packages        3.2 kB/s | 5.0 kB      00:01
Package openssh-server-9.9p1-9.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost ~]#
```

```
systemctl start sshd
```

Step 3: Enable SSH at Boot

```
systemctl enable sshd
```

Step 4: Check Status

```
systemctl status sshd
```

```

[root@localhost ~]# systemctl enable sshd
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: ena>
   Active: active (running) since Fri 2026-07-17 10:51:08 IST; 3min 25s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 1000 (sshd)
    Tasks: 1 (limit: 4273)
  Memory: 128.0K (peak: 2.3M)
     CPU: 19ms
    CGroup: /system.slice/sshd.service
            └─1000 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 17 10:51:07 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jul 17 10:51:08 localhost.localdomain sshd[1000]: Server listening on 0.0.0.0 p>
Jul 17 10:51:08 localhost.localdomain sshd[1000]: Server listening on :: port 2>
Jul 17 10:51:08 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)

```

**Step
5: Test
SSH
to**

localhost

ssh localhost

Are you sure you want to continue connecting (yes/no)?

Type:

yes

Enter your password.

Verify login.

Exit:

exit

2.Generate a new SSH key pair using ssh-keygen, and add your public key to the ~/.ssh/authorized_keys file of your user account to enable key-based authentication.

Create SSH Key Authentication

Step 1: Switch to your normal user

su – du

Step 2: Generate SSH Key

ssh-keygen

Press **Enter** for all prompts.

Example:

Enter file in which to save the key:

Press **Enter**

Enter passphrase:

Press **Enter**

Enter same passphrase again:

Press **Enter**

Step 3: Verify Keys

ls
~/ssh

```
[du@localhost ~]$ ssh-keygen -t rsa
Generating public/private rsa key pair.
Enter file in which to save the key (/home/du/.ssh/id_rsa):
Created directory '/home/du/.ssh'.
Enter passphrase for "/home/du/.ssh/id_rsa" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/du/.ssh/id_rsa
Your public key has been saved in /home/du/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:mMNdVot8du++K/MgargUrRQ/wknqqy7ai/3Wb5lu50Q du@localhost.localdomain
The key's randomart image is:
+---[RSA 3072]-----+
|
| . o .
| o = + .
| .++=o o . .
| .=*S+E .
| . ..+.. .
| o o.o.. .
| = . +.=oo .o..
|+ **O. *=+. +++
+-----[SHA256]-----+
[du@localhost ~]$
```

Step 4: Copy Public Key

```
cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_key
```

Step 5: Set Permissions

```
chmod 700 ~/.ssh
```

```
chmod 600 ~/.ssh/authorized_key
```

Step 6: Verify

```
ls -l ~/.ssh
```

```
[du@localhost ~]$ ls -l ~/.ssh
total 12
-rw-----. 1 du du 578 Jul 17 11:05 authorized_key
-rw-----. 1 du du 2610 Jul 17 11:01 id_rsa
-rw-r--r--. 1 du du 578 Jul 17 11:01 id_rsa.pub
[du@localhost ~]$
```

3. Edit the /etc/ssh/sshd_config file to disable password authentication by setting PasswordAuthentication no, then restart the sshd service and test that you can only log in using your SSH key.

Disable Password Login

Switch back to root

exit

Step 1: Edit SSH Configuration

vi /etc/ssh/sshd_config

Find:

#PasswordAuthentication yes

Change it to:

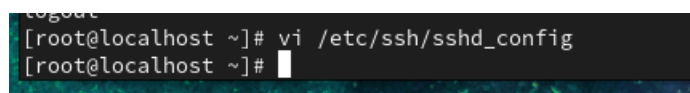
PasswordAuthentication no

If it already exists as yes, change it to no.

Save:

Esc

:wq

A terminal window screenshot showing the command 'vi /etc/ssh/sshd_config' being executed. The prompt is '[root@localhost ~]#'. The background of the terminal has a dark, abstract pattern.

Step 2: Restart

SSH

systemctl restart sshd

Step 3: Verify

systemctl status sshd

```
[root@localhost ~]# systemctl restart sshd
[root@localhost ~]# systemctl status sshd
● sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: ena>
   Active: active (running) since Fri 2026-07-17 11:09:48 IST; 10s ago
     Docs: man:sshd(8)
           man:sshd_config(5)
  Main PID: 7324 (sshd)
    Tasks: 1 (limit: 4273)
   Memory: 1.3M (peak: 1.6M)
      CPU: 22ms
   CGroup: /system.slice/ssh.service
           └─7324 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Jul 17 11:09:48 localhost.localdomain systemd[1]: Starting OpenSSH server daemo>
Jul 17 11:09:48 localhost.localdomain sshd[7324]: Server listening on 0.0.0.0 p>
Jul 17 11:09:48 localhost.localdomain sshd[7324]: Server listening on :: port 2>
Jul 17 11:09:48 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
lines 1-16/16 (END)
```

Step 4:
Test
Key
Login
Switch
to user:
su - du
Run:

ssh localhost

If configured correctly, SSH should log in using the key without asking for the user's password.

Exit:

exit

```
[du@localhost ~]$ exit
logout
Connection to localhost closed.
```

**4.Document each step you took to configure SSH key-based authentication and disable password login, including the exact commands and configuration changes you made.

Hint: Use screenshots or command outputs to support your documentation.**

Documentation

```
195 dnf install openssh-server -y
196 systemctl start sshd
197 systemctl enable sshd
198 systemctl status sshd
199 ssh localhost
200 exit
201 su - du
202 vi /etc/ssh/sshd_config
203 systemctl restart sshd
204 systemctl status sshd
205 su - du
206 history
[root@localhost ~]#
```

```
71 ssh-keygen
72 ssh-keygen
73 ls ~/.ssh
74 ls ~/.ssh/id_ed25519
75 whoami
76 ssh-keygen -t rsa
77 ls -la ~/.ssh
78 ls ~/.ssh
79 cat ~/.ssh/id_rsa.pub >> ~/.ssh/authorized_key
80 chmod 700 ~/.ssh
81 chmod 600 ~/.ssh/authorized_keys
82 chmod 600 ~/.ssh/authorized_key
83 ls -l ~/.ssh
84 exit
85 ssh localhost
86 history
du@localhost ~]$
```